5 A small conducting sphere of mass of 0.12 g is suspended from a very long insulating thread between metal plates M and N, separated by 2.0 cm as shown in Figure 5.1.

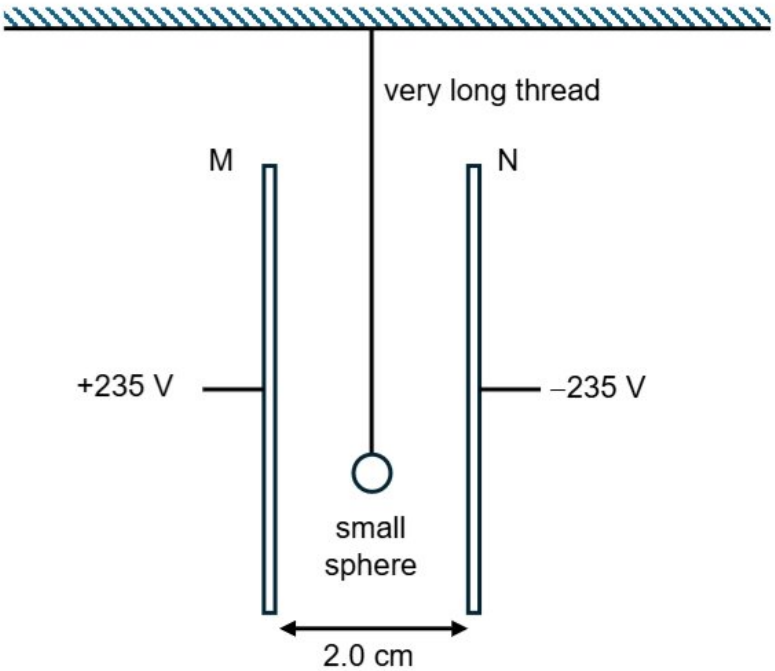


Figure 5.1 (not to scale)

Plates M and N are connected to a high-voltage power supply and are held at potentials of 235 V and -235 V, respectively.

After being given an initial displacement, the sphere moves to plate M, touches it and acquires a charge of 58 pC.

It then moves towards plate N, touches it and the horizontal oscillation process repeats itself.

Assume that the initial velocity of the sphere is negligible when moving away from the plates.

(a) Calculate the electrostatic force on the sphere as it moves towards plate N.

force = ... N [2]

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(b) Calculate the acceleration of the sphere.

acceleration = ... m s^{-2} [1]

(c) Determine the frequency of oscillation of the sphere. Assume the size of the sphere is negligible compared to the separation of the plates.

frequency = ... Hz [3]

(d) A student claims that the oscillation is simple harmonic.

Explain whether the student is correct.

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[Total: 7]